Nanomedicine

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***Abstract*—Designing nanoparticles that target tumors without causing any major side effects is a major challenge that requires slow, expensive lab testing. This project investigates using machine learning (ML) to streamline that process. We implemented machine learning models, specifically Random Forest, Deep Neural Networks & XGBoost on nanoparticle data to predict Biodistribution. These models were then integrated with multi-objective optimizer models such as NSGA(Non-dominated Sorting Genetic Algorithm) - III, to find the best design parameters. By using the ML models as fitness functions, the algorithm was able to identify Pareto Front of nanoparticle configurations. This allowed us to simultaneously maximize tumor accumulation while minimizing side effects like off-target toxicity to organs like the liver and spleen. Our results demonstrate that this framework can be used to predict optimal nanoparticle designs before any physical experiments are conducted, significantly reducing the time and cost of drug delviery research.**

I. Introduction

Nanomedicine looks very promising for targeted drug delivery by transporting therapeutic agents to diseased tissues while sparing the healthy organs of any side effects. However, achieving precise biodistribution remains a major challenge. The biological fate of a nanoparticle is determined by a complex interplay of its composition (size, shape surface charge etc), making manual optimization highly inefficient. Traditional methods struggle to keep up with the complexity of these interactions, thus needing automated, data-driven approaches.

Current research increasingly utilizes ML to analyze these patterns. While previous studies have focused on forward prediction, calculating the outcome of a known particle, very few have addressed the "inverse design" problem: determining which particle properties are required to achieve a specific therapeutic result. Our research improves upon existing methodologies by combining high-accuracy predictive models with multi-objective optimization. We propose a novel pipeline that not only predicts biodistribution but also prescribes the optimal physicochemical features needed to balance safety and efficacy.

II. Methodology

The proposed system for optimizing nanoparticle design follows a pipeline-based workflow consisting of 3 main stages: data preprocessing, surrogate model development, and the integration of the multi-objective optimization engine within a user interface (UI). The system is designed to take raw experimental data and transform it into a tool that can prescribe the best nanoparticle designs based on specific user constraints.

A. Pre-processing and Feature Extraction

The first component of the methodology serves to standardize and transform raw experimental data into a form suitable for machine learning models. Standardizing this data is essential to ensure that the models can accurately capture the relationships between a nanoparticle's physical properties and its biological behavior.

- **Iterative Data Refinement** Throughout the development process, we adopted an iterative approach to data preprocessing. Because nanoparticle-biological interactions are highly complex, and human errors occur, we continuously identified and corrected inconsistencies in the raw experimental logs, such as measurement noise and outlier values that initially skewed model predictions.
- **Data Cleaning and Standardization** We began by formatting the dataset to remove structural noise and handle missing values in the nanoparticle characteristic logs. Continuous variables, such as hydrodynamic diameter and zeta potential **(have to double check this)** were standardized to ensure that each feature contributes equally during the training progress.
- **Handling Class Imbalance** A common issue in biodistribution data is class imbalance. To ensure fair training, we applied random under-sampling to create a balanced dataset, ensuring that each biodistribution category, such as high tumor uptake versus low tumor uptake was represented equally.

B. Forward Model Architecture

The second component of our system is the development of a **"Forward Model"** which acts as a virtual laboratory. We implemented and compared multiple deep learning architectures to capture the non-linear relationships in the data.

- **Random Forest (RF)**
  The RF model serves as a baseline using a multi-output regression strategy. Specifically, an independent Random Forest regressor was trained for each target organ (e.g., Tumor, Heart, Liver). To address missing data points, a masked training approach was utilized, where the loss function ignored *NaN* values for specific organ concentrations during the bootstrap aggregation process.

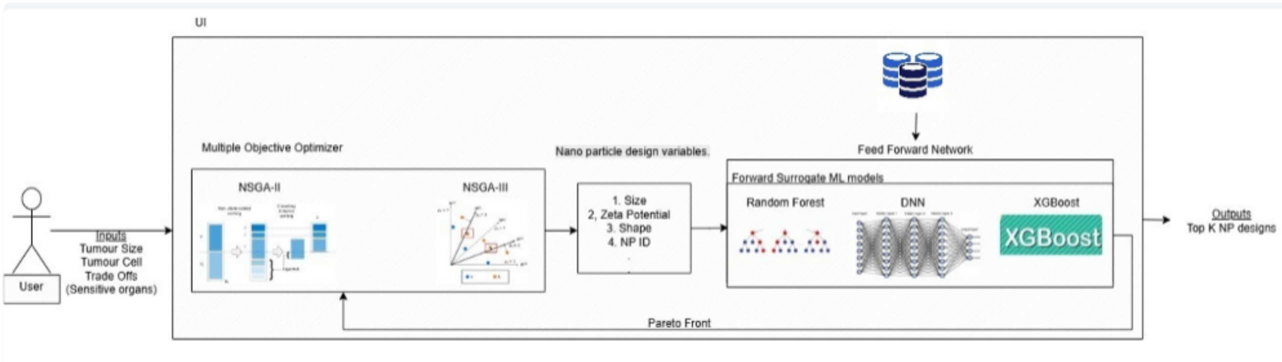

Fig. 1: Proposed Nanoparticle Optimization Pipeline.

- **Deep Neural Networks (DNN)**
  To capture non-linear relationships in heterogeneous data, a Deep Neural Network (DNN) was implemented using the **pytorch-tabular** framework. This architecture utilizes a **Category Embedding** layer, which maps categorical design parameters into a continuous latent space. The model was trained using a custom **Masked Mean Squared Error** loss to account for partially labeled samples across the six target organs.
- **Extreme Gradient Boosting (XGBoost)**
  The third architecture employs a hierarchical approach using Extreme Gradient Boosting (XGBoost). This model functions as a **Specialist Ensemble**, where input data is routed to specific sub-models based on the tumor cell type.
  - **Data Transformation:** Numerical features, including dose and time, were log-transformed ($y = \log(1+x)$) to normalize the distribution and stabilize gradient updates.
  - **Specialization:** By training specific models for different tumor cell lines, the architecture accounts for the high variability in nanoparticle-cell interactions.

C. System Integration and Inverse Design

The final stage of the methodology integrates the forward surrogate models into a functional framework that enables inverse design. This system is designed to transition from simple forward prediction to a prescriptive tool that suggests optimal nanoparticle configurations based on specific user constraints.

- Closed-Loop Optimization Logic.
  The core of the integration is a closed-loop interaction between the predictive machine learning models and the multi-objective optimization engine.
  1) **Forward Model Integration**: The trained forward model serve as the evaluation function within the optimization loop. For each candidate nanoparticle design generated by the optimizer, a corresponding feature vector is constructed and passed to the surrogate models. These models predict biodistribution across multiple organs, including tumor accumulation and off-target uptake. The predicted values are used directly as objective functions, where tumor accumulation is maximized while accumulation in healthy organs is minimized.
  2) **Multi-Objective Optimization**: A multi-objective optimization framework based on NSGA-III is employed to explore the design space. Each candidate solution consists of both continuous variables (e.g., size, zeta potential, PDI) and categorical variables (e.g., core material, targeting strategy, shape). The optimizer evolves a population of solutions to approximate the Pareto front representing trade-offs between targeting efficiency and toxicity.
  3) **Constraint-Aware Design**: Explicit constraints are incorporated to ensure physically and biologically feasible nanoparticle designs. These include (i) physical consistency constraints such as hydrodynamic size being greater than or equal to core size and bounded PDI values, (ii) biological constraints such as a minimum tumor accumulation threshold, and (iii) strategy-dependent constraints where the feasible ranges of size and surface charge depend on whether passive or active targeting is selected.
  4) **Mixed-Variable Representation**: The optimization problem includes both continuous and categorical variables. Categorical features are encoded as indexed variables and mapped back to their respective categories during evaluation, enabling simultaneous optimization of structural design choices and physicochemical properties.
  5) **Iterative Closed-Loop Process**: The system operates iteratively, where the optimizer proposes candidate designs, the surrogate models evaluate their performance, constraint violations are penalized, and the population is updated based on Pareto dominance and diversity preservation. This process continues until convergence.

6) **Design Ranking and Selection**: The resulting Pareto-optimal solutions are ranked using a selectivity metric defined as the ratio of tumor accumulation to weighted accumulation in healthy organs. This allows identification of nanoparticle designs that achieve high targeting efficiency with reduced systemic toxicity.

D. Interactive User Interface (Nanoparticle Portal)

To facilitate the practical application of the inverse design framework, a web-based research portal was developed using the **Streamlit** framework. This interface serves as a high-level abstraction layer, allowing researchers to interact with the NSGA-III optimization engine without direct code manipulation.

- **Scenario Customization:** The portal allows users to define specific biological contexts, including tumor cell type, tumor volume (cm^3), and tumour body weight (g). These inputs are utilized as fixed context variables in the forward prediction models during the optimization loop.
- **Dynamic Toxicity Weighting:** A key innovation in the UI is the *Organ Sensitivity* module. Researchers can assign individual penalty weights (w_i) to healthy organs. These weights are used to calculate a **Selectivity Score** (S), defined as:

$$S = \frac{C_{tumor}}{\sum(C_{healthy,i} \cdot w_i) / \sum w_i} \tag{1}$$

where C represents the predicted concentration (%ID/g). This allows for personalized optimization based on specific clinical safety profiles.
- **Constraint-Aware Guardrails:** The interface enforces biological and physical feasibility by applying hard constraints during the search process, such as ensuring the Polydispersity Index (PDI) ≤ 0.2 and maintaining a minimum tumor accumulation threshold of 10%.
- **Data Visualization and Export:** The portal provides real-time visual feedback through interactive Pareto trade-off plots and predicted biodistribution profiles. Optimized nanoparticle "blueprints" can be exported as CSV files, streamlining the transition from computational design to physical synthesis.

E. Hyperparameter Configuration

All models underwent rigorous tuning using Bayesian Optimization via the Optuna framework. The final configurations are summarized in Table I.

TABLE I: Model Hyperparameter Settings

Model	**Key Hyperparameter**	**Value/Range**
Random Forest	Estimators (n_{trees}) Random State	100 42
DNN-CE	Loss Function Framework	Masked MSE PyTorch Tabular
XGBoost	Learning Rate (η) Max Depth Tree Method	0.01 – 0.2 3 – 9 Hist (GPU)

III. Results

A. Performance Overview

The research produced a comprehensive set of predictive and optimization outcomes. By utilizing over 2,300 experi-

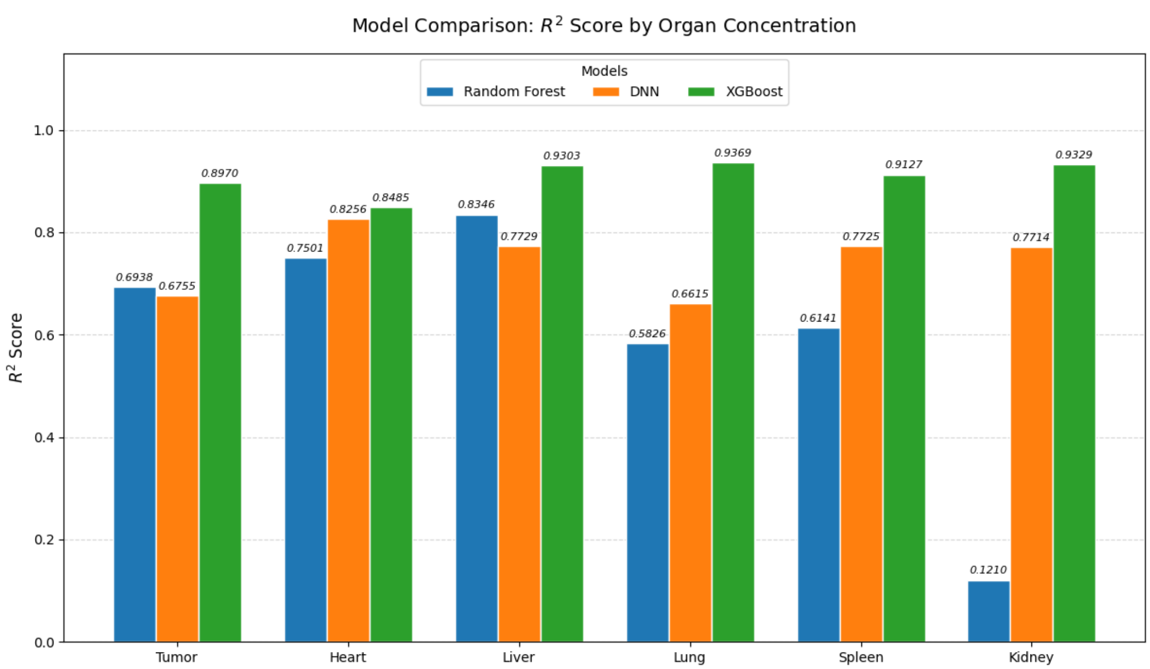

Fig. 2: Comparative Analysis of Forward Model Performance across target organs.

mental data points, the forward models established a mathematical link between nanoparticle physical blueprints and their biodistribution. These predictions were then used as the foundation for an inverse design process that identified optimal design candidates.

B. Evaluation Framework and Metrics

To rigorously assess predictive reliability, a multi-metric approach was applied to a held-out 20% test set. These metrics account for the inherent biological variability and the sparse nature of the experimental data:

- **Coefficient of Determination** (R^2)**:** Measures the proportion of variance in organ concentration captured by the model.
- **Mean Absolute Error (MAE):** Represents the average magnitude of prediction errors in %ID/g (percent injected dose per gram).
- **Root Mean Squared Error (RMSE):** Penalizes larger outliers to ensure consistency across different dosage levels.
- **Masked Evaluation:** As established in the methodology, a masked loss logic was used during testing to ensure that missing data points did not skew the accuracy results.

C. Forward Model Validation and Comparison

The performance of the three implemented architectures: Random Forest, DNN-CE, and Specialist XGBoost was benchmarked across the primary biological compartments. As shown in Figure 2, the Specialist XGBoost model consistently demonstrated higher R^2 scores across critical organs.

The Specialist XGBoost architecture achieved high precision in the Liver ($R^2 = 0.9303$) and Kidney ($R^2 = 0.9329$), validating the effectiveness of routing data to tumor-specific specialists.

D. Model Reliability and Correlation Analysis

To further assess the robustness of these predictions, a reliability assessment was conducted comparing true versus predicted values.

As illustrated in Figure 3, predictions cluster tightly along the ideal fit line for primary clearance organs. Furthermore, feature importance analysis revealed that physical properties such as size and zeta potential are the primary drivers of biodistribution.

E. Feature Importance and Global Impact

Following the correlation analysis, the global impact of individual design and context variables was quantified using the XGBoost feature importance scores. This analysis provides a granular view of which parameters drive the virtual lab predictions across all organ targets.

As shown in Figure 4, physical properties such as **Zeta Potential** and **Size (HD)** emerged as the most dominant predictors of nanoparticle behavior. Interestingly, experimental context variables like **Time Point** and **Dose** also showed significant importance, highlighting that biodistribution is a time-dependent biological process.

F. Inverse Design Analysis and Pareto Optimization

The insights from the forward model were utilized to conduct an inverse design analysis, mapping the relationship between predicted concentrations in competing organs.

Figure 5 demonstrates the distribution of candidate designs across the organ concentration space. The clustering suggests specific design zones where high tumor accumulation can

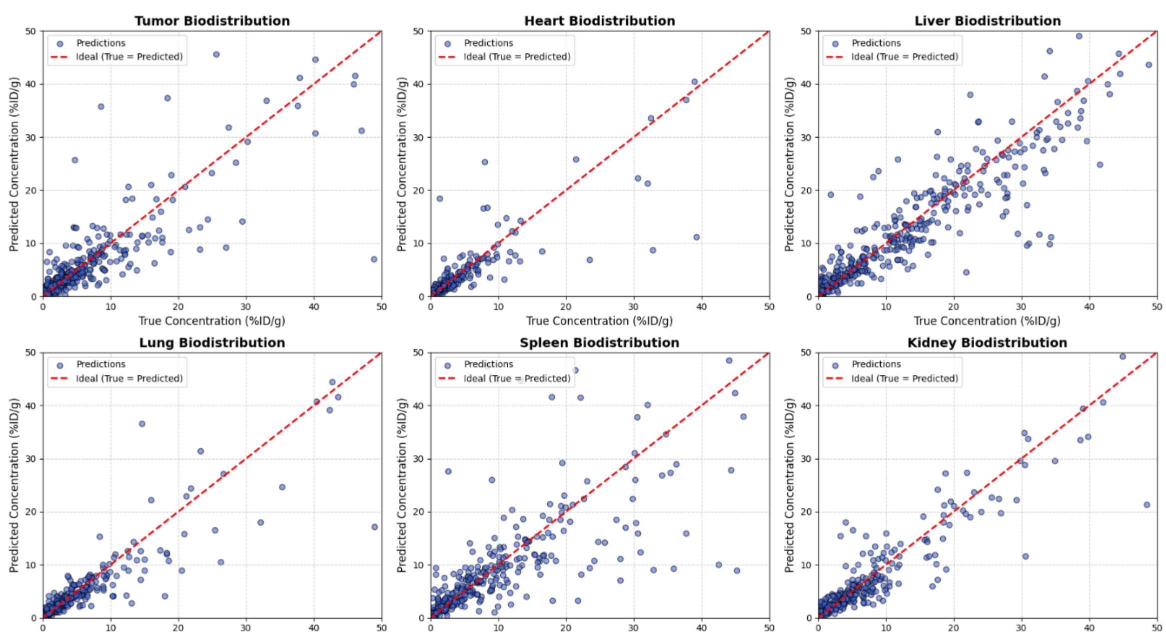

Fig. 3: Model Reliability Assessment: Global correlation across all organs.

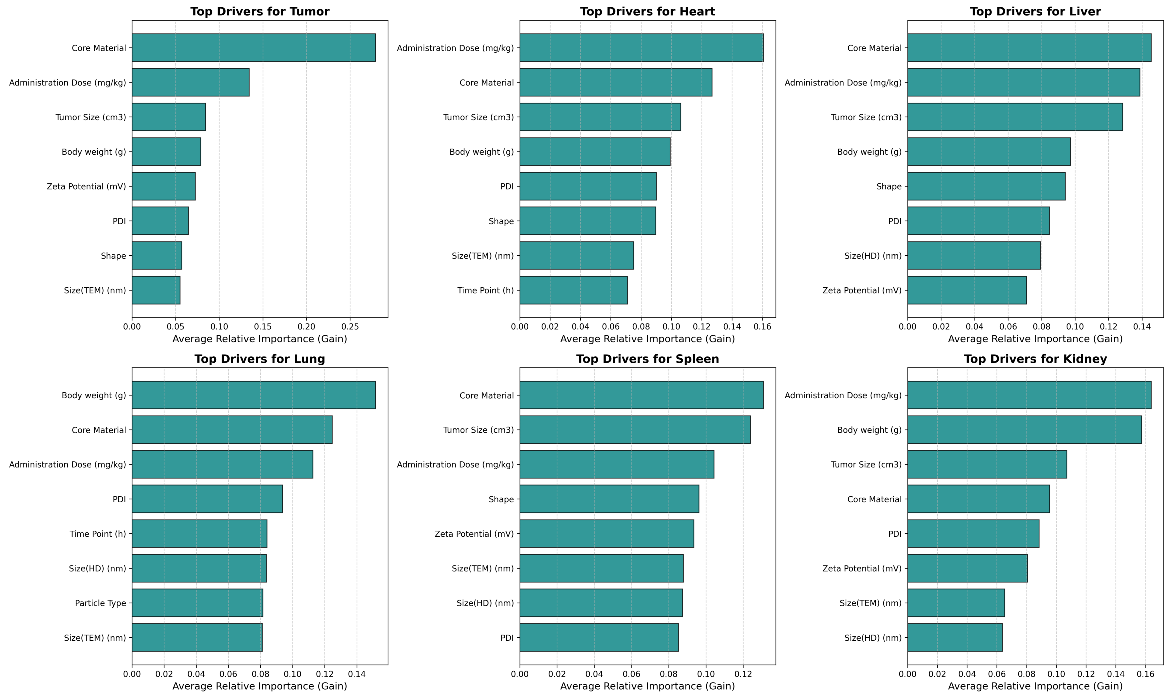

Fig. 4: Combined Feature Importance Scores for Design and Context Variables.

be achieved without a proportional increase in healthy organ toxicity. To finalize the selection, the NSGA-III optimizer identified the Pareto front, representing the mathematical limit of the design trade-offs.

The Pareto front illustrated in Figure 6 reveals the inherent trade-offs between competing biological objectives. Candidates on this front represent the most efficient configurations, where tumor targeting cannot be improved without increasing off-target accumulation.

G. Prescriptive Blueprints and Optimal Design Selection

The final output of the proposed framework is a set of actionable "blueprints" for nanoparticle synthesis. By applying the selectivity metric defined in the **Interactive User Interface (Nanoparticle Portal)** (Section II - D) , the system identifies specific physicochemical configurations from the Pareto front that represent the most viable candidates for laboratory validation.

These top-performing candidates are detailed in **Table II**, which lists the precise structural and experimental parameters prescribed by the optimizer. As shown in the table, the Rank #0 design achieved the highest overall Selectivity Score of 6.49.

The predicted biological performance of these top designs is visualized in **Figure 7**. These charts demonstrate a consistent ability to maintain high tumor accumulation while significantly suppressing concentrations in off-target organs such as the heart and lungs.

A critical analysis of the blueprints in **Table II** reveals several key trends:

- **Structural Preferences:** The optimizer consistently favored *carbon nanotubes* and *rod* morphologies. This aligns with the **Feature Importance and Global Impact** analysis in Section III-E (Feature Importance and Global Impact), where these high-aspect-ratio structures were associated with superior biodistribution profiles.ctures were associated with superior biodistribution profiles.
- **Consistency with Constraints:** All designs in **Table II** adhere strictly to the physical guardrails defined in the methodology, such as maintaining a PDI ≤ 0.2 for high-scoring active targeting candidates and satisfying the 10% minimum tumor accumulation threshold.

TABLE II: Top 10 Optimized Nanoparticle Design Blueprints

Rank	Size(TEM)	Size(HD)	Zeta (mV)	PDI	Dose	Time	Core Material	Strategy	Shape	Score
0	174.28	6.49	62.49	1.30	12.96	29.95	carbon_nanotube	active	rod	6.49
1	180.42	290.18	61.76	1.29	10.17	20.85	carbon_nanotube	passive	rod	4.85
2	171.61	248.37	61.51	1.45	18.50	21.03	carbon_nanotube	passive	rod	4.69
3	205.11	302.21	59.84	1.33	10.69	56.95	carbon_nanotube	passive	rod	4.67
4	204.84	11.60	51.21	0.90	12.92	63.11	carbon_nanotube	active	rod	4.54
5	207.84	294.65	63.94	1.67	16.39	59.49	anticancer drug	passive	rod	4.35
6	213.36	17.40	46.95	1.66	9.84	24.38	carbon_nanotube	active	rod	4.28
7	214.10	3.15	63.16	1.35	661.59	27.00	anticancer drug	active	rod	4.13
8	187.88	283.99	39.29	1.27	95.76	32.03	anticancer drug	passive	rod	4.09
9	170.67	247.71	60.99	1.49	18.15	34.48	anticancer drug	active	rod	4.07

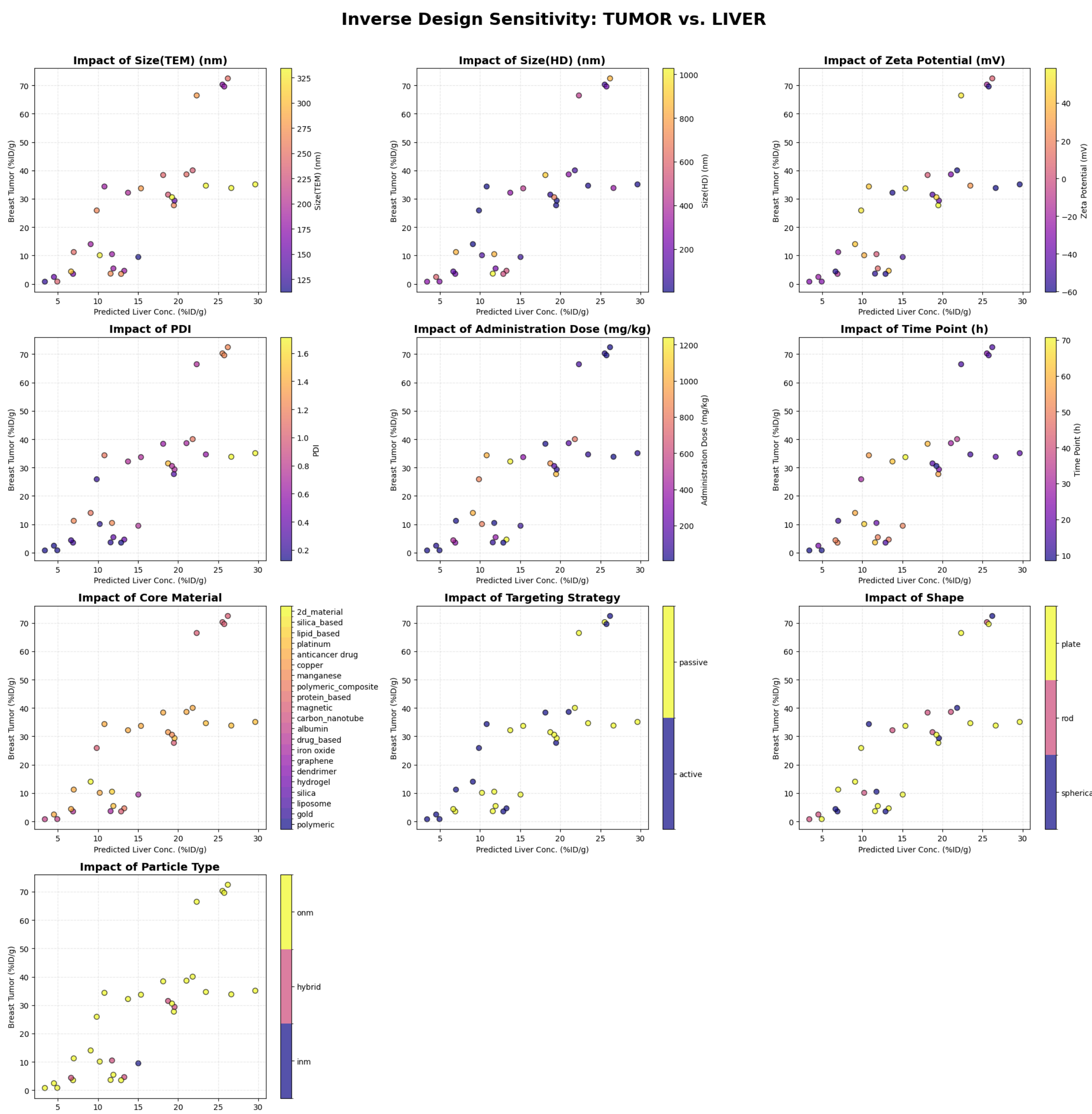

Fig. 5: Inverse Design Analysis showing the distribution of candidate nanoparticle configurations across competing organ concentration spaces.

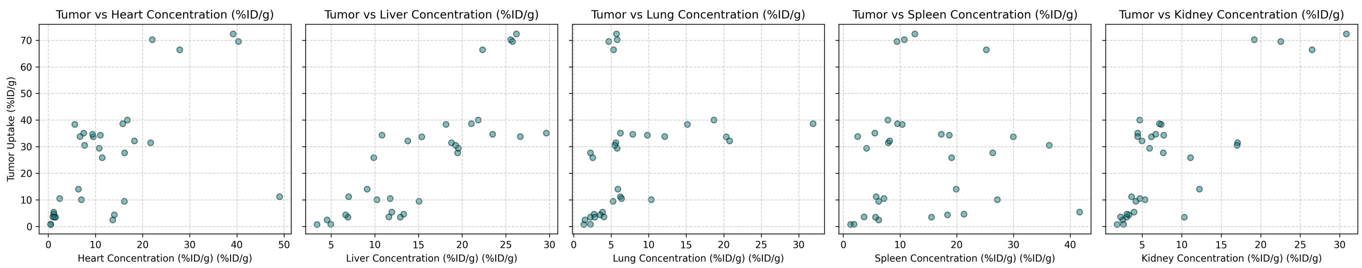

Fig. 6: Pareto Front Visualization for Multi-Objective Optimization showing the mathematical limit of the design trade-offs.

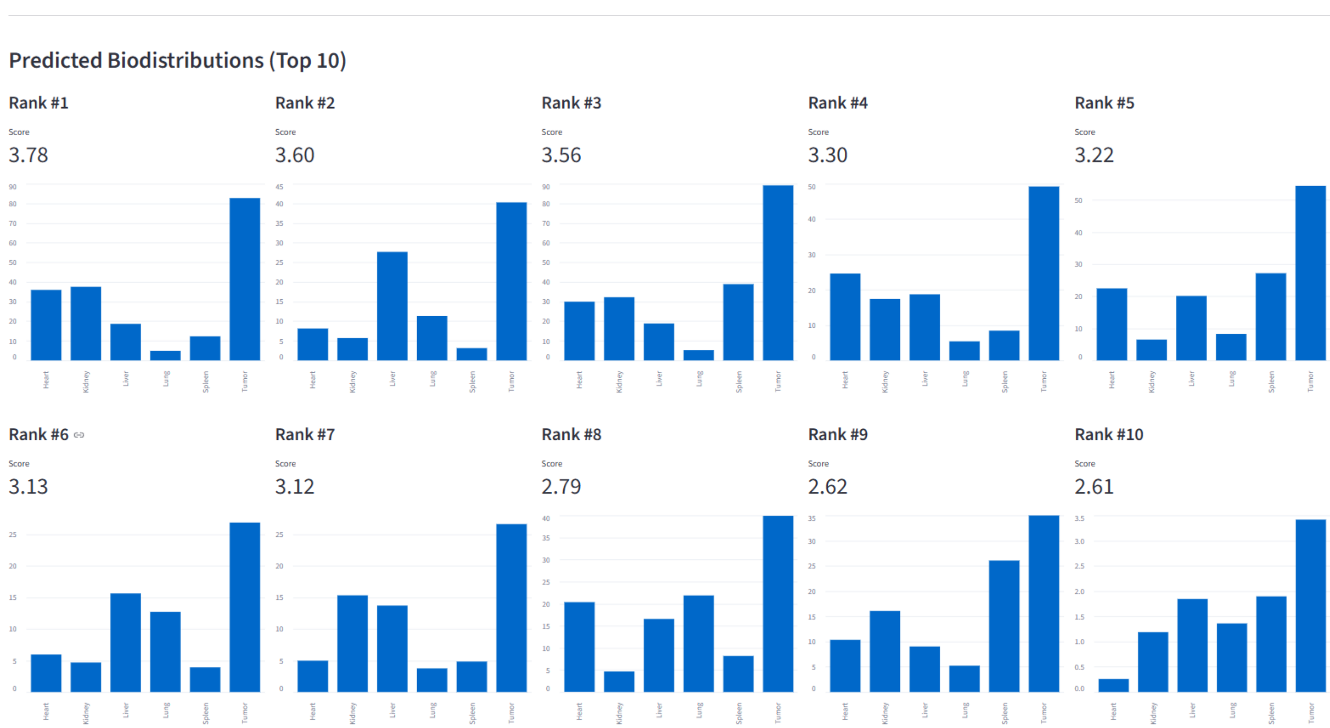

Fig. 7: Predicted Organ Accumulation for the Top 4 Optimized Designs. The ranking is based on the Selectivity Score, which balances high tumor targeting against minimal off-target uptake in clearance organs.

IV. Conclusion

This research establishes a data-driven framework for the inverse design of nanoparticles, successfully bridging the gap between predictive machine learning and prescriptive structural engineering. By integrating specialist XGBoost architectures with multi-objective NSGA-III optimization, we have demonstrated that it is possible to mathematically derive nanoparticle "blueprints" that maximize therapeutic efficacy while strictly adhering to safety and physical feasibility constraints. The development of the interactive Nanoparticle Portal further democratizes these complex models, providing researchers with a functional tool to personalize optimization based on specific clinical toxicity concerns. This approach builds upon existing meta-analyses of nanoparticle distribution in tumor-bearing models [1], offering a closed-loop pipeline to accelerate the development of targeted nanomedicines and reduce the reliance on costly, iterative trial-and-error laboratory experimentation.

References

[1] Q. Chen, L. Yuan, W. C. Chou, and Z. Lin, "Meta-Analysis of Nanoparticle Distribution in Tumors and Major Organs in Tumor-Bearing Mice," *ACS Nano*, vol. 17, no. 20, pp. 19810–19831, 2023.